

Strengthening and weakening Addressee Compliance by constraining discourse future: the case of Mandarin *ba*

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Abstract

This paper investigates how opposing discourse effects of either weakening or strengthening the sentential force can be derived from constraining projected discourse future, through a case study of the Mandarin discourse particle *ba*. *ba* has a weakening effect in declarative and imperative uses, but a strengthening effect in morphologically marked interrogatives. Previous literature has either treated the two types of uses of *ba* as distinct particles, or fails to provide a unified account across different clause types. This paper argues that *ba* conveys a single conventional meaning: it signals potential or existing addressee non-compliance in discourse. The core idea is that *ba* presupposes an epistemic asymmetry between speaker and addressee, and contributes a discourse effect that keeps the projected common ground updates unchanged from input to output. In declaratives, this blocked projection of addressee acceptance reflects an assumption that the addressee may not comply, yielding a weakening effect. In interrogatives, failing to project any possible answers signals a conversational crisis: the question itself seems unanswerable. Restating a seemingly unanswerable question with *ba* strengthens its interrogative force. Implemented within a Farkas & Bruce (2010)-style Table model, this account also successfully explains the weakening effect of *ba* in imperatives. The paper also contributes to formal models of discourse dynamics, in particular the role of its still poorly-understood projected components.

Keywords: Discourse particles, Table model, clause types, semantics-pragmatics, Mandarin

1 Introduction

Interlocutors soften their tone to express uncertainty or politeness, and strengthen it to convey impatience or anger. In Mandarin, both effects can be achieved with sentence-final discourse particles, and sometimes even by the same one. The particle examined in this paper, *ba*, gives rise to a weakening effect in declaratives, but a strengthening effect in interrogatives, as illustrated in (1).

(1) a. **Weakening in a declarative**

Xiayu-le **ba**.
rain-PFV BA
'(I think) it rained.'

b. **Strengthening in an interrogative**

Ni qu-bu-qu **ba**?!
2SG GO-NEG.IMP-GO BA
'Are you going or not (I am losing my patience now)?! '

The contrasting sentences in (1) exemplify the central puzzle to be addressed in this paper: how can *ba*'s seemingly opposite effects across clause types be derived from a unified mechanism conventionally encoded by a single discourse particle?

Recent formal literature on discourse provides a pathway towards a unified treatment of different clause types. The influential Farkas and Bruce (2010) Table model elaborates the Stalnakerian update of assertions by adding an intermediate step in the context update process: before updating the Common Ground, the content of an utterance is first *proposed* (i.e., placed onto the Table). It is then the addressee's responsibility to move the conversation forward, either by accepting the proposal with linguistic signals or gestures (e.g., nodding), or rejecting the proposal with explicit linguistic cues such as the English polarity particle *no*. The Common Ground is updated only when the content is accepted by all interlocutors. In this way, the Table model unifies the discourse effects of assertions and questions in terms of what Farkas (2022, 2024) more recently termed Addressee Compliance: not only canonical questions but assertions too presume that the addressee will respond with certain default discourse moves as in (2).¹

(2) a. **Addressee compliance for canonical questions:**

Speaker presents herself as assuming that Addressee will resolve [the question...] in the immediate future of the conversation.

¹Farkas (2024) originally formulates the assumptions for assertions and questions separately, as in (2). The two assumptions can be unified if we assume both assertions and questions raise *issues*, as stated below.

(1) **Addressee compliance:**

Speaker assumes that Addressee will commit to a proposition in the Issue that has been raised.

In the case of assertions, there is only one proposition in the Issue, and the addressee is expected to either accept or reject that proposition. In the case of questions, the speaker assumes that the addressee will commit to a potential answer to the question. The main difference in the question case concerns the requirement of immediacy ("...the immediate future of the conversation"), which is not captured by (2) but is also crucial for the discussion of interrogatives later in §5. I thank Deniz Rudin (p.c.) for drawing my attention to this point.

b. **Addressee compliance for canonical assertions:**

Speaker assumes that Addressee will accept [the proposal] in the future of the conversation.

Non-canonical assertions and questions are thus defined as suspending the default assumptions for their canonical counterparts.² For example, Farkas (2022) shows that Romanian particle *oare* forms a special type of non-canonical question called *non-intrusive* as a result of suspending Addressee Compliance for questions.

Following the insights of Farkas (2022, 2024), the weakening and strengthening effects that *ba* adds can therefore be characterized in terms of Addressee Compliance: in declaratives, *ba* weakens Addressee Compliance; in interrogatives, it strengthens it. However, that description stays at the level of the actual pragmatic effects *ba* gives rise to. The crucial question is whether these seemingly opposite effects can be derived from a single underlying mechanism. To foreshadow a bit, in the rest of the paper, I argue that Addressee Compliance corresponds to the default projection operation in the Table model, whereby canonical future discourse moves are projected. On this view, *ba*'s apparently opposite discourse effects follow from a single mechanism that constrains the context in which such compliant discourse futures are projected. Specifically, I propose that *ba* presupposes that the speaker is not assumed to be epistemically privileged, thereby blocking the projection of a compliant discourse future. The next section introduces the basic facts about the particle to present in detail the core phenomena the paper aims to capture.

1.1 *ba*

ba is an unembeddable sentence-final particle documented in declaratives and morphosyntactically marked interrogatives (e.g., Chao 1968). Note that in Mandarin, declaratives and imperatives can almost never be morphosyntactically marked. When declarative sentences are used for directive speech acts, they are sometimes called imperatives (e.g., Ettinger and Malamud 2015). (An exception is the dedicated prohibitive marker *bie* in matrix clauses, see Yip 2016; Liao and Wang 2022; Wimmer 2025 for related discussion). For this reason, when a *ba*-marked utterance receives an imperative interpretation, it will occasionally still be referred to as a *ba*-declarative in this paper. When *ba* attaches to declaratives it typically serves to add a weakening flavor to the prejacent. For instance, the *ba*-attached utterance in (3b) conveys an additional 'uncertainty' inference compared with (3a).

- (3) a. Sue ganmao-le.
Sue have.a.cold-PFV
'Sue has a cold.'
- b. Sue ganmao-le **ba**.
Sue have.a.cold-PFV BA
'(I think) Sue has a cold.'

²Farkas (2024) provides complete sets of assumptions for assertions and questions. I omit the rest and focus on Addressee Compliance for the current paper.

ba-interrogatives, by contrast, typically appear in ‘impatience’ scenarios (Ettinger and Malamud, 2015), expressing the speaker’s frustration or anger. In (4) and (5), *ba* adds an additional ‘unfriendly’ flavor to the questions: the speaker is impatient about the current conversation status and wants the addressee to give an answer immediately. The underlined parts of (4) and (5) signals that these are syntactically-marked interrogatives: *shenme* is a *wh*-word, meaning ‘what’; and *chi-bu-chi* is an A-not-A construction, marking one type of polar question in Mandarin.

- (4) Ni daodi yao shenme **ba**?
 you on-earth want what BA
 ‘(Tell me), what on earth do you want?’ (Chao 1968: 807)
- (5) Daodi chi-bu-chi dangao **ba**?
 on-earth eat-NEG-eat cake BA
 ‘Are you going to eat cake (I am losing my patience now!)?’

The function of *ba* in declaratives is usually characterized as some sort of weakening effect in the literature (e.g., “soliciting-agreement” (Li and Thompson 1989), “weakening the neustic” (Han 1995), “speaker’s uncertainty of the propositional content” (Chu 2009)). The interrogative use of *ba* has traditionally been treated as a distinct particle from declarative *ba* (Chao, 1968: 810). More recently, Ettinger and Malamud (2015) try to provide a unified account of the meaning of *ba* as marking a contingent move. I consider the core idea underlying their proposal to be essentially correct, as it captures the intuition that the function of *ba* is related to addressee responsibility. However, Ettinger and Malamud (2015)’s discourse model incorrectly predicts that *ba* cannot attach to syntactically-marked interrogatives. Although they discuss some cases of *ba*-interrogatives and correctly point out that *ba*-interrogatives are only felicitous in very restricted discourse contexts (i.e., the ‘impatience’ scenarios), they fail to integrate the so-called ‘impatience’ condition into an overall account of the semantics and pragmatics of *ba*.

Weakening effects on declarative force have also been analyzed as partial or absent speaker commitment, as in English rising declaratives (e.g., Gunlogson 2004; Rudin 2018, 2022), or a lowered degree of speaker credence in the available evidence, as in English tag questions (e.g., Farkas and Roelofsen 2017). §2 will show that *ba* differs from both: it conveys strong speaker commitment, and sometimes may occur when speaker(s) have high credence in evidence (e.g., when the evidence is direct).

The rest of this paper will scrutinize the distribution of *ba* in declaratives, interrogatives, and imperatives, presenting analyses for them implemented in a Lewisian scoreboard model. The common core connecting the uses of *ba* in different clause types is that *ba* signals either a potentially or an existing non-compliant move from the addressee by constraining future discourse developments, i.e., how an addressee would react to a *ba*-attached utterance. More specifically, *ba* does not project any potential updates on the common ground, which has opposite discourse effects in declaratives and interrogatives. In declaratives, the softening effect of *ba* can be understood as weakening the Addressee Compliance assumption for canonical assertions. I challenge two existing accounts of the source of *ba*’s weakening effect: the question-particle view (e.g., Zhu 1982) and the epistemic-modal view (e.g., Xu 2003).³ I present novel evidence for *ba* conveying that the speaker is uncertain about whether

³A similar position has previously been advocated by Deng (2015). However, the primary concern of Deng (2015) is syntactic, namely the syntax–pragmatics mapping of *ba*. By contrast, the present paper sets aside syntactic issues and focuses exclusively on the discourse effects of *ba*.

the utterance will be accepted by the addressee. This kind of speaker uncertainty is formulated as the (pragmatic) presupposition induced by *ba*, which requires the interlocutors not to assume that the speaker is epistemically more privileged. In interrogatives, the ‘impatience’ flavor can be understood as a strengthening of the Addressee Compliance assumption: a *ba*-interrogative urges the addressee to answer the question in the immediate future of the conversation. This type of scenario typically occurs when the addressee withholds an answer to a question she is committed to addressing, or when the question appears unanswerable, leading to a potential conversational crisis. In imperatives, *ba* weakens Addressee Compliance by being felicitous only in contexts where the imperative does not impose an obligation on the addressee to comply.

The rest of the paper is structured as follows. §2 argues that, contrary to prior analyses, *ba* is neither a question particle nor an epistemic modal, instead presenting evidence that *ba* is sensitive to speaker (a)symmetry in knowledge. §3 develops an account of *ba* in declaratives that explains their softening effect. §4 establishes the (in-)felicitous contexts where *ba*-interrogatives may or may not occur. Following on these contextual limitations, §5 derives the unfriendly effect of *ba* in interrogatives by positing specific presuppositions on the input context. §6 discusses the implications of the proposed theory on imperative uses of *ba*. §7 concludes.

2 *ba* in declaratives

This section lays out the empirical patterns of *ba*-declaratives.⁴ Three properties will be illustrated: (i) *ba*-declaratives are inherently assertive; (ii) *ba* presupposes the speaker is not mutually believed to be an epistemic authority in the preceding context; and (iii) *ba* is not an epistemic modal. *ba*’s contextual restriction on the (a)symmetry in knowledge naturally gives rise to a particular kind of uncertainty: a speaker is uncertain about whether her utterance will be accepted, given that she is not believed to be the authority. I further distinguish this from the uncertainty expressed by epistemic modals, and argue that *ba* does not necessarily express epistemic uncertainty (contrary to Li and Thompson 1989; Xu 2003; Chu 2009 *i.a.*).

2.1 The assertive potential

ba-declaratives possess assertive potential and do not function as genuine questions. In Mandarin, it is quite common to form a question by simply adding a sentence-final particle, such as the polar question particle *ma* (Chao 1968; Li and Thompson 1989; Li 2006; Bhatt and Dayal 2020; Yuan and Hara 2021; Dayal 2023; Liu and Luo 2025 *i.a.*). *ba* is sometimes described in the literature as a question-forming particle (Zhu 1982), or as marking the ‘interrogative mood’, (Chao 1968). I show that *ba*-utterances with declarative prejacent are not genuine questions by adopting the so-called ‘tell me’ test (Asher and Reese 2007; Kamali 2020): utterances that follow ‘tell me’ must carry at least a questioning component. As shown in (6), a *ba*-utterance with a declarative prejacent is infelicitous following *gaosuwo* ‘tell me’. The same prejacent, when combined with the question particle *ma* or structured in the *A-not-A* construction, can naturally co-occur with ‘tell me’, as shown in (7). This contrast provides evidence that *ba*-declaratives lack interrogative force.

⁴Throughout this section, I use *ba*-declaratives to refer to *ba*-utterances whose prejacent are not morphosyntactically marked as interrogatives.

- (6) #Gaosu wo, ni mingtian qu xuexiao **ba** .
 tell 1SG 2SG tomorrow go school BA
 Intended: ‘Tell me, will you go to school tomorrow?’
- (7) a. Gaosu wo, ni mingtian qu xuexiao **ma**?
 tell 1SG 2SG tomorrow go school SFP
 b. Gaosu wo, ni mingtian **qu-bu-qu** xuexiao?
 tell 1SG 2SG tomorrow go-NEG-go school
 ‘Tell me, will you go to school tomorrow?’

In addition, an utterance’s assertive component can be tested with the ‘after all’ test (Asher and Reese 2007): an assertion, or a complex speech act that is partially assertive, is compatible with utterances of ‘after all’, while a pure questioning speech act is not. The utterance with *ba* survives the ‘after all’ test as in (8). By contrast, a question marked by *ma* or structured in the *A-not-A* construction is not acceptable following *bijing* ‘after all’, as shown in (9).

- (8) Bijing, ni mingtian yao qu xuexiao **ba** .
 after.all 2SG tomorrow will go school BA
 ‘After all, you will go to school tomorrow.’
- (9) a. #Bijing, ni mingtian qu xuexiao **ma**?
 after.all 2SG tomorrow go school SFP
 b. #Bijing, ni mingtian **qu-bu-qu** xuexiao?
 after.all 2SG tomorrow go-NEG-go school
 ‘After all, will you go to school tomorrow?’

The second set of diagnostics is drawn from Rudin (2022) and is applied by Liu and Luo (2025) to test the interrogativity of *ba*-declaratives. Rudin (2022) observes that while canonical assertions allow the addressee to raise a related but new issue as a possible continuation, canonical questions do not license such responses unless the question has first been addressed. (10) demonstrates that assertions may have three types of potential continuations: acceptance (10a), rejection (10b), and a new issue (10c).⁵ By contrast, in response to a question, a new issue cannot be raised without addressing the question first, as in (11).

- (10) [Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she has a new girlfriend. He turns to Bertha, who is close with Carrie, and says:]
 A: Carrie got a new girlfriend. **assertion**
 a. B: Yeah, she told me about it this morning. (acceptance)
 b. B: I don’t think so. Maybe she’s just trying to stir up drama. (rejection)
 c. B: %Did you know Delia is leaving her husband? (new issue)
- (11) A: Did Carrie get a new girlfriend? **question**
 a. B: Yeah, she told me about it this morning. (positive answer)

⁵The judgment of (10c) is mine. For Rudin (2022), (10c) is fully acceptable.

- b. B: I don't think so. Maybe she's just trying to stir up drama. (negative answer)
 - c. B: #Did you know Delia is leaving her husband? (new issue)
- (Rudin 2022: 345)

Liu and Luo (2025) applied Rudin's test to Mandarin *ba*-declaratives and *ma*-questions; an A-not-A question is added here. Only *ba*-declaratives—and not *ma*-attached or A-not-A questions—allow 'raising a new issue' as a felicitous continuation, as shown in (12) and (13).

- (12) A: Carrie jiao-le ge xin nvpengyou **ba**.
 Carrie get-PFV CL new girlfriend BA
- a. B: Queshi. 'Indeed.' (acceptance)
 - b. B: Wo tingshuo Delia ye jiao-le ge xin nvpengyou.
 I hear Delia also get-PFV CL new girlfriend
 'I heard that Delia also got a new girlfriend.' (new issue)
- (13) A: Carrie jiao-le ge xin nvpengyou **ma?**/ Carrie **jiao-mei-jiao** ge xin nvpengyou?
 Carrie get-PFV CL new girlfriend MA Carrie get-NEG.PFV-get CL new girlfriend
- a. B: Queshi. 'Indeed.' (positive answer)
 - b. B: #Wo tingshuo Delia ye jiao-le ge xin nvpengyou.
 I hear Delia also get-PFV CL new girlfriend
 'I heard that Delia also got a new girlfriend.' (new issue)
- (Liu and Luo 2025: 29)

I conclude that *ba*-declaratives are inherently assertive and do not contain a questioning speech act component. The following subsection highlights the pragmatic presupposition that *ba* imposes on the input context: the speaker of a *ba*-utterance is not mutually believed to be epistemically more privileged concerning the issue under discussion. While I will elaborate further in §3, the following discussion will occasionally address how *ba*'s sensitivity to epistemic (a)symmetry between interlocutors can be understood as a type of speaker uncertainty: specifically the speaker's uncertainty about whether the asserted content will be accepted by the addressee.

2.2 (A)symmetry in knowledge

ba presupposes it is not common ground the speaker is epistemically more privileged. When it is mutually believed that the speaker holds epistemic authority over the addressee, the use of *ba* becomes infelicitous, as illustrated in (14), whereas a canonical assertion is considered the most appropriate response.

- (14) [A has been working in a windowless room, and B has just arrived.]
 A: *What is the weather like now?*
- a. B: # Xiayu le **ba**.
 rain PFV BA
 - b. B: Xiayu le.
 rain PFV
 'It's raining.'

Four typical scenarios where *ba* may naturally occur are exemplified below. The four scenarios are constructed to vary according to whether the speaker holds epistemic authority over the addressee. I show that *ba* is felicitous in all cases *except* when it is public belief that the speaker clearly holds epistemic authority over the addressee, as illustrated in (14).

Confirmation-seeking. When the two interlocutors are in a situation where it is public belief that the addressee knows better about the issue under discussion, the speaker may use a *ba*-declarative to confirm. In the scenario in (15), an implicit question ‘when is the party?’ is assumed. The speaker has a rough idea of when the party starts but expects the addressee, as the organizer, to have more accurate information. The particle *ba* in this declarative indicates that the speaker is uncertain about whether the addressee agrees with her, as the addressee holds greater epistemic authority over the issue.

(15) CONFIRMATION SEEKING

[The addressee is organizing a party. The speaker is wondering when exactly the party starts.]

Jinwan paidui shi qidian **ba**.
tonight party COP 7-o’-clock BA

‘The party is at 7 o’ clock tonight, right?’

At first glance, the confirmation-seeking context is similar to an English *rising declarative*, a sentence with a rising intonational contour that is morphosyntactically declarative but also has a questioning speech act component (Gunlogson 2004, 2008; Truckenbrodt et al. 2008; Malamud and Stephenson 2015; Farkas and Roelofsen 2017; Westera 2017; Rudin 2018, 2019, 2022 *i.a.*). However, the *ba*-declarative in (15) should not be taken as carrying a questioning component since it conveys a strong speaker commitment to the content of the prejacent. If the speaker entertains an alternative possibility, as in (16a), the presence of a second *ba* makes the utterance sound contradictory (see also Yang 2021). In contrast, the same utterance is perfectly acceptable within a sequence of two modalized sentences as in (16b), or plain declaratives with question intonation represented by ↑, as in (16c).

- (16) a. Jinwan paidui shi qidian **ba**. #Huozhe shi badian **ba**.
tonight party COP 7-o’-clock BA or COP 8-o’-clock BA

Intended: ‘The party is at 7 o’ clock tonight? Or at 8?’

(*ba*: ✗)

- b. Jinwan paidui **keneng/yinggai** shi qidian. Huozhe **keneng/yinggai** shi
tonight party probably/should COP 7-o’-clock or probably/should COP
badian.
8-o’-clock

‘The party is probably/should be at 7 o’ clock tonight. Or It is probably/ should be at 8.’

(modals: ✓)

- c. Jinwan paidui shi qidian↑? Huozhe shi badian↑?
tonight party COP 7-o’-clock or COP 8-o’-clock

‘The party is at 7 o’ clock tonight? Or at 8?’

(question intonation: ✓)

Broadly speaking, English rising declaratives fall into two types: (i) *assertive* rising declaratives, which convey full speaker commitment to the content of the prejacent, with any uncertainty being metalinguistic; and (ii) *inquisitive* rising declaratives, which do not incur speaker commitment (Jeong

2018).⁶ An inquisitive rising declarative may convey opposite speaker biases depending on the context: either a bias in favor of the prejacent (the context in 17a) or a bias against it (in 17b). Mandarin declaratives with question intonation are felicitous in both cases, while *ba*-declaratives are only acceptable in (17a), but not (17b).

(17) a. **Expressing positive bias:**

[A saw B sneezing and guessed that B might have a cold, though it could also be an allergic reaction.]

A: Ganmao-le↑? /Ganmao-le **ba**.

get.a.cold-PFV get.a.cold-PFV BA

‘(You) got a cold?’

b. **Expressing distrust:**

[C saw B sneezing and guessed that B might have a cold, but A knows that B has allergies and often sneezes.]

C: *B might have a cold.*

A: Ganmao-le↑?/ #**ba**. Bu keneng, kending shi guomin fanying

have.a.cold-PFV BA NEG.IMPF possible must COP allergy reaction

‘B has a cold? That can’t be, it must be an allergic reaction.’

Confirmation-seeking *ba* utterances are often translated into English as tag questions.⁷ However, there are also cases where a tag question is perfectly acceptable but a *ba*-declarative is not, as exemplified in (18) (see also Liu and Luo 2025 for related discussion).

- (18) [Amalia and Bert have a colleague, Carla, who is admired by everyone for her problem solving skills. One day, at a department meeting, the chair raises an issue, and Carla immediately finds

⁶Jeong (2018) identifies five distinct uses of rising declaratives in English, as illustrated in (1). In her analysis, (1a–c) are classified as inquisitive uses, while (1d–e) are assertive uses. This paper adopts this classification.

(1) a. **Contradictory Questions**

A: Please apologize to him.

B: *I was wrong and I should apologize?* No way.

b. **Incredulous Questions**

A: John went to the airport to pick up his sister.

B: *John has a sister?*

c. **Confirmative Questions**

A: (airline agent): There’s one flight to Helsinki.

B: (customer): *The flight leaves at 4pm?*

d. **Unsure about a metalinguistic issue**

A: Do you speak Spanish?

B: *I speak Ladino?*

e. **Uptalk**

(A waitress in a restaurant talking to a customer)

B: Hello! *My name is Anna? I’ll be your waitress?*

(Jeong 2018: 307)

⁷For recent development on English tag questions, see Tottie and Hoffmann (2006); Asher and Reese (2007); Malamud and Stephenson (2015); Krifka (2015); Farkas and Roelofsen (2017); Wiltschko et al. (2018); Frana and Rawlins (2019); Scheffler and Malamud (2023) among many others.

a solution that makes everyone happy. Amalia to Bert:.]

Amalia: She always finds a solution that makes everyone happy, doesn't she?

Amalia: #Ta zong neng zhaodao yi-ge rang suoyouren manyi-de fangan **ba**
3SG always can find one-CL make everyone happy-REL solution BA

(Farkas and Roelofsen 2017: 36)

Quiz question. *ba* may occur in answers to quiz questions, as shown in (19). It is assumed that Speaker A holds greater epistemic authority over Speaker B in this context, as B is the one being tested.

(19) QUIZ QUESTION:

[In an oral exam:]

A: *What is the capital of France?*

B: Bali **ba**.

Paris BA

‘(I think it’s) Paris.’

Addressee-uncertainty. *ba* can be used to answer a question to which the addressee does not know the answer either. This is shown by the SUE’S ENGLISH case in (20).

(20) SUE’S ENGLISH:

A: *How’s Sue’s English?*

B: Ting hao de **ba**.

very good BA

‘(Should be) pretty good.’

When A asks a genuinely information-seeking question, B is assumed to be more epistemically privileged with respect to Sue’s English ability. However, this is merely A’s assumption or private belief; by using *ba*, B backs away from those assumed commitments. The difference between (20) and the infelicitous context (14) is whether both interlocutors agree on speaker’s authority status: in (14) it is public belief that B knows whether it is raining given the contextual settings, but in (20) the context does not make it clear whether B also thinks she knows about Sue’s English.

Note that B may or may not be confident in her assessment of Sue’s English ability. When B is confident in her statement, the use of *ba* generates a politeness effect. The felicity condition described so far concerns only the public beliefs (or public commitments, a distinction not drawn in this paper), rather than whether the speaker actually holds such authority in the context. It is generally assumed that a speaker will not commit herself to not knowing something if she in fact does know it. However, she may do so when there is a need to pretend to be ignorant or to be polite, and example (20) may illustrate such a case. While this paper does not discuss cases involving pretense, it is worth keeping them in mind, as such usages show that *ba* is not about speaker’s actual epistemic state, but what the speaker acts like (or commits herself to).

Disagreements. Lastly, when both interlocutors hold opposing views on an issue and have access to evidence of comparable strength, *ba* can be used to express disagreement with the other interlocutor, as shown in (21).

(21) *A: It's raining outside, I saw someone carrying a wet umbrella just now.*

B: Mei xia **ba**, wo ye gang dao, gangcai mei xiayu.
 NEG.PFV rain BA 1SG also just arrive just.now NEG.PFV rain

'It should not be, I also just arrived and it wasn't raining.'

It is worth noting that the use of *ba* is comparable to *should* in (20) and (21). Here, both *should* and *ba* imply some kind of reasoning and cannot be used if the speaker is known to know the true answer. Consider (22), where both A and B directly witness the raining event. It is infelicitous to use *yinggai* 'should' or *ba* when the contextual evidence is direct, but for different reasons: *yinggai* is incompatible with situations where the speaker already knows the content of the utterance, while *ba* is unacceptable when the speaker is highly certain that the utterance will be accepted (e.g., when both A and B have access to the same direct evidence).

(22) [A and B directly see that it's raining. A to B:]

A: # Yinggai xiayu le.
 should rain PFV

A': # Xiayu le **ba**.
 rain PFV BA

Intended: 'It should be raining.'

The indirectness constraint is not necessary for a *ba*-declarative to be felicitous if the speaker has reason to believe that her utterance might not be accepted. In a suitable context, what A says in (22) can be felicitous with direct evidence. In (23), the utterance *it's raining* does not appear discourse-initially; A and B have been arguing about whether it's raining prior to the *ba*-utterance. This example again shows that *ba* is not about knowledge, but about beliefs about knowledge: the content of the prejacent can be already known to both interlocutors, as long as there is no epistemic asymmetry between the two.⁸

(23) *A: It's raining outside, I saw someone carrying a wet umbrella just now.*

B: It can't be, I arrived 5 minutes ago and it wasn't raining.

[A and B go outside and directly see that it's raining. A to B:]

A: Xiayu le **ba**
 rain PFV BA

'(As I've told you,) it's raining!'

⁸For *ba* to be felicitous in (23), a special intonation on *ba* is required for some native speakers. For discussions on the effects of intonation/stress on discourse particles, see Grosz (2020) for German particles *ja/doch*, McCready (2006); Davis (2009); Oshima (2014) among many others for Japanese sentence-final particle *yo*, and Yuan and Hara (2019) for Mandarin *ma*.

2.3 *ba* is not a bona fide epistemic modal

Although *should* and *ba* share commonalities in their meaning, I argue that *ba* is not an epistemic modal itself. First, *ba* and epistemic modals differ in their embedding properties: *ba* is an unembeddable, utterance-final particle, whereas epistemic modals can be embedded in at least some contexts. In (24) we test whether embedding *ba* and *yinggai* ‘should’ under attitude verbs is possible or not. Note that in contrast to the modal *yinggai*, which is interpreted as depending on the attitude encoded by the matrix predicate (see Yalcin 2007’s proposal for English epistemic modals), *ba* takes scope over the attitude report and expresses overall uncertainty towards this very report of Zhangsan’s thinking.

- (24) a. Zhangsan juede waimian **yinggai** xiayu le.
 Zhangsan think outside should rain PERF
 ‘Zhangsan thinks that it should be raining outside.’
 b. A: Why is Zhangsan wearing a raincoat today?
 B: Zhangsan juede waimian xiayu le **ba**.
 Zhangsan think outside rain PERF BA
 ‘(I think it’s because) Zhangsan thinks that it’s raining outside.’

Furthermore, *ba* does not survive the *should p, but not p* test in (25), whereas Mandarin *yinggai* ‘should’ can be followed by the negation of its prejacent. Note that when *dan ta bu shi* ‘but it isn’t’ is absent in (25b), the sentence *the beer is cold by now* is perfect with *ba*, which has the inferential evidential meaning: given the evidence that the speaker has access to, the speaker infers that *p*.

- (25) a. The beer should be cold by now, but it isn’t.
 b. # Pijiu dao xianzai weizhi shi lengde le **ba**, dan ta bu shi.
 beer by.now COP cold PFV BA but it NEG COP
 Intended: ‘The beer should be cold by now, but it isn’t.’ (ba: ✗)
 c. Pijiu dao xianzai weizhi **yinggai** shi lengde le, dan ta bu shi.
 beer by.now should COP cold PFV but it NEG COP
 ‘The beer should be cold by now, but it isn’t.’ (yinggai: ✓)
 (modelled after Okano and Mori 2014’s Japanese example)

When the speaker makes an adversarial utterance such as *you are an idiot*, the use of *ba* is felicitous as in (26a), as the speaker assumes the addressee will not accept it.⁹ In contrast, the use of epistemic modals in such contexts is infelicitous as in (26b).

- (26) [A gave B a sandwich to eat.]
 A: By the way, the sandwich you just ate had been in the fridge for five months.
 a. B: Ni shi baichi **ba**!
 2SG COP idiot BA
 ‘You are an idiot!’

⁹(26) conveys an inference strengthened from ‘not assuming addressee’s acceptance’ to ‘assuming addressee’s rejection’. The aim of such an utterance may be just to offend the addressee as in (26), or to publicize an unpopular belief as in (21).

- b. B: # Ni yinggai/keneng/kending shi baichi !
 2SG should/might/must COP idiot
 ‘You should/might/must be an idiot!’

Lastly, *ba* may also indicate speaker uncertainty of whether the answer they provide resolves the question, whereas modals cannot, as shown in (27).¹⁰

- (27) [Assume that B knows for sure about her sister’s car color].

A: *Is your sister’s car some crazy color?*

B: Tade che shi juhongsede **ba**...
 her car COP orange BA

‘Her car is orange.....(but I don’t know if it’s crazy).’

B’: # Tade che yinggai/kending/keneng shi juhongsede...
 her car should/must/might COP orange

‘Her car ought to be orange...’

To sum up, we have seen that *ba*-declaratives are assertoric in nature. The felicity of *ba* in a declarative reflects the (a)symmetry of epistemic authority between the interlocutors. A *ba*-declarative can be used when the addressee knows better (e.g., confirmation-seeking or quiz question scenarios), when both interlocutors are equally informed (as in the disagreement scenarios), or when it is not obvious who knows better (as in the SUE’S ENGLISH scenario). I further compared *ba* with epistemic modals and suggest that although *ba* can display an epistemic modal-like meaning, it does not systematically behave like a modal. In the next section, I will show how (a)symmetry in epistemic authority explains the weakening effect triggered by *ba* and develop an account of *ba* that captures these properties.

3 *ba* weakens Addressee Compliance in declaratives

In this section, I propose that *ba* weakens the Addressee Compliance assumption in declaratives, and that this special weakening effect is connected to an epistemic (a)symmetry condition imposed by *ba*, as discussed in §2.2. The proposal is implemented in a Farkas & Bruce-style Table model (§3.3), and relevant theoretical assumptions are introduced in §3.2.

3.1 From (a)symmetry in authority to weakening Addressee Compliance

§2.2 showed that using *ba* is infelicitous when it is mutually believed that the speaker is in an epistemically more privileged position with respect to the issue under discussion (as in 14). I dub this the

¹⁰See Constant (2014) for similar uses of lone contrastive topics (or sentential CTs), where the CT contour is added to the whole sentence instead of one phrase. Lone CTs also carry the inference that the speaker is uncertain if an answer fully answers an open question, as shown in (1).

- (1) A: Did anything interesting happen today?
 B: [Persephone came over]_{CT}...

(Constant 2014: 24)

Anti-Authority Condition in (28), and propose that the Anti-Authority Condition should be treated as a *pragmatic presupposition* in the sense of Karttunen (1974) (defined in 29).

(28) **Anti-Authority Condition:**¹¹

With a prejacent property *P*, *ba* presupposes that it is *not* common ground that the speaker is in a more epistemically privileged position concerning *P*.

(29) **Pragmatic presupposition:**

Sentence A pragmatically presupposes proposition B iff it is the case that A can be felicitously uttered only in contexts which entail B.

(Karttunen, 1974: 149).

When the Anti-Authority Condition is met, only a subset of declarative utterances is felicitous. The speaker of a declarative utterance may have epistemic privilege over the issue under discussion. Cases like (14) where the speaker is taken to have adequate evidence fall into this category. But this is not true for all declaratives. Plain declaratives are also felicitous in contexts such as the quiz context (19) or the disagreement context (21), where the speaker is not assumed to have such authority. In these contexts, *ba*-declaratives are felicitous. In other words, what *ba* contributes is the explicit exclusion of input contexts in which the speaker of a declarative utterance would be mutually believed to know more. Consider also the sentence in (30). In any unmarked context, using *ba* is infelicitous in (30) since the speaker is considered an authority on what languages she can speak.

- (30) Wo hui shuo yingyu { $\emptyset$ / #*ba*} .
 1sg can speak English BA
 ‘I can speak English.’

It is debatable whether the speech acts performed in these contexts are genuine assertions. Alternatively, they could be characterized as *guesses* (in quiz contexts) or *predictions* (in disagreement contexts, Mandelkern and Dorst 2022). On a weak assertion view, as argued by Mandelkern and Dorst (2022), declaratives in guessing contexts are assertions. Others argue that declaratives made in guessing and predictive contexts are weaker speech acts (e.g., Benton and Turri 2014; Ninan 2025). While the actual debate is orthogonal to this paper, we can safely maintain that *ba*-declaratives are weaker uses, or at least constitute non-canonical assertive speech acts.

The restriction on epistemic authority in the input context also influences the overall discourse effects of a *ba*-declarative on future discourse developments. The idea is the following: if in a given context it is not presupposed that the speaker holds epistemic authority over the addressee, then the speaker should not assume that the addressee will accept *p* as true by default. This overall effect can be alternatively stated using the notions introduced in §1: If we take Addressee Compliance for canonical assertions (31) to consist in an assumption that the addressee will accept what the speaker says as true, then what *ba* contributes is a weakening of that assumption. This amounts to say that *ba* signals awareness of a potentially non-compliant move from the addressee. The Addressee Compliance assumption is inherently Gricean. Following Grice’s maxim of quality (i.e., *be truthful*; Grice 1975), a cooperative speaker is assumed to express what she believes to be true, and a cooperative addressee is correspondingly assumed to take what the speaker expresses as true. When the speaker is believed to

¹¹ The Anti-Authority Condition will be naturally satisfied in a genuine information-seeking question.

occupy an epistemically privileged position, the addressee has no reason to challenge or doubt what the speaker asserts (the *master/slave* scenario, Lewis 1979). However, when the epistemic asymmetry between speaker and addressee is more balanced or even flipped, as presupposed by *ba*, it is reasonable to expect that the addressee may cast doubt on or even reject the speaker's assertions.

(31) **Addressee compliance for canonical assertions:**

Speaker assumes that Addressee will accept *p* in the future of the conversation.

(Farkas 2024: 10)

The connection between epistemic authority and the discourse effects of certain clause types has also been explored in an attempt to account for the discourse effects of imperatives. To explain the self-verifying effect of imperatives, Kaufmann (2012) proposes that (at least stereotypical) imperatives presuppose that the speaker occupies a privileged position, such that “the addressee will accept as true the speaker's utterances about what she commands the addressee to do” (Kaufmann 2012: 147). In other words, stereotypical imperatives exclude all contexts in which the speaker lacks epistemic privilege, whereas *ba* confines *ba*-attached declaratives to precisely such contexts. We will discuss what predictions this proposal makes about the imperative uses of *ba* in §6.

3.2 Theoretical background

The discourse model adopted here is a Lewisian ‘conversational scoreboard’, which keeps track of discourse moves and contextual information exchange, representing the conversational state in the course of conversation (Lewis 1979). The original Lewisian scoreboard has been enriched and further developed, by Farkas and Bruce (2010), Malamud and Stephenson (2015), and Farkas and Roelofsen (2017). I assume the scoreboard components from Farkas and Bruce (2010), in which the public commitments that each interlocutor makes can be traced, and the possible resolutions of the current issue are reflected as the *projected set*. The paper will adopt the notations of contextual components used by Farkas and Roelofsen (2017) and Farkas (2024), in which both declarative and interrogative sentences are construed as *issues*, and hence receive the same semantic type (a set of propositions).

(32) An issue *I* is a non-empty set of propositions.

In this way, the denotation of a declarative or an interrogative sentence is always a set of sets of possible worlds. A declarative sentence thus differs from an interrogative only in how many propositions the issue raised by the sentence contains. If an issue only contains a single proposition (i.e., it is a singleton set), then it is raised by a declarative; if it contains at least two propositions, then it is an interrogative.

So far, we have seen that declaratives and interrogatives are treated uniformly. Bearing this in mind, the context structure components this paper uses to characterize *ba* are specified below.

(33) Context $c = \langle A, T, DC_x, CG, PS \rangle$, where

- a. Participants (*A*): a set of discourse participants (*sp* for speaker, *a* for addressee);
- b. The Table (*T*): a stack of sets of propositions (issues) (Ginzburg, 1996; Roberts, 1996; Büring, 2003; Farkas and Bruce, 2010);
- c. Discourse Commitments (DC_x): the set of propositions that each discourse participant is publicly committed to (Gunlogson, 2004, 2008);

- d. Common Ground (CG): Set of propositions that all discourse participants mutually agree on being true as a basis for joint action (Stalnaker, 1978; AnderBois, 2014; Faller, 2019);
- e. Projected Set (ps): the set of supersets of the current CG, each of which is a projection of a future CG that might be used to decide the topmost issue on the Table (Farkas and Bruce, 2010).

The component in the context that I rely on most for my analysis is the Table. The Table is defined as a stack of issues (sets of propositions), which keeps track of the *proposals* (the QUDs in Ginzburg 1996 and Roberts 1996) for updating the common ground (Farkas and Bruce, 2010). In a normal course of conversation, interlocutors are expected to ‘steer’ the conversation toward increasing the common ground, i.e., to resolve what is on top of the stack. The Table will be ‘emptied’ once the current issue gets resolved. The Common Ground is updated with the content only when the content is accepted by all interlocutors.

Apart from the Table, there are three other conversational components that also play crucial roles in my account: the Discourse Commitments sets (DC_x) for each interlocutor x , and the Projected Set (ps). The DC_x used in my account is adapted from the definition of *public beliefs* proposed by Gunlogson (2004): a proposition p is a public belief of A (is a member of A ’s commitment set) if and only if the proposition ‘ A believes p ’ is in the common ground. In other words, a public belief of an individual is not necessarily a mutual belief, and propositions in DC_x are not necessarily part of the Common Ground. In this way, a Stalnakerian common ground can be achieved by taking the intersection of the interlocutors’ commitment sets.

In order to show the different effects a canonical assertion or a question may have on the contexts, Farkas and Bruce (2010) propose that canonical speech acts by default project a set of future common grounds, the Projected Set (ps). ps suggests potential ways of resolving the current issue. When a canonical assertion is put on the Table, ps will project that the addressee accepts the assertive content. When a canonical question is raised, ps will be updated with all the possible answers to the question Q . The updating operation $ps \sqcup P$ is defined in (34), which says that an updated ps is a new collection of possible developments of the common ground, and each future *cg* is created by adding one proposition in P to the previous *cg*. The future *cgs* in the new collection ps should be consistent; inconsistent future *cgs* will be eliminated.

(34) Definition of ps:¹²

- a. Let $ps = \{cg_1, \dots, cg_n\}$ be a set of sets of propositions (e.g. possible common grounds) and let $P = \{p_1, \dots, p_m\}$ be a set of propositions.
- b. $ps \sqcup P = \{cg_i \cup \{p_j\} \mid cg_i \in ps \text{ and } p_j \in P \text{ and } \bigcap (cg_i \cup \{p_j\}) \neq \emptyset\}$
(i.e. only keeps the consistent future common grounds)

(Modified from Farkas and Bruce 2010)

Under this framework, Farkas and Bruce’s speech act operators (assertion and question operator) can be merged as the following basic *conventional discourse effects* (CDE) since declarative and interrogative sentences have the same semantic type (Farkas and Roelofsen 2017; for effect by clause type or logical type, see also Kaufmann 2012; Lauer and Condoravdi 2012; Condoravdi and Lauer 2012; Portner 2004). CDEs are functions from input contexts (K_i) to output contexts (K_o) determined by a

¹²By (33e), $CG \subseteq cg_i$, for all $1 \leq i \leq n$.

particular sentence form. Only updated discourse components are listed below; unmentioned aspects remain the same as their inputs. The subscripts i and o stand for input and output respectively.

(35) Basic CDES of updating a sentence S (content: I):

$$K_i + S = K_o:$$

- a. $DC_{sp,o} = DC_{sp,i} \cup \bigcup I$
- b. $T_o = \text{PUSH}(I, T_i)$
- c. $PS_o = PS_i \cup I$

(35) says that when an issue I is raised by a sentence S , two basic updates are made to input contexts: first, $\bigcup I$ is added to the speaker's commitment set. When I contains only a single proposition (e.g., in a declarative with denotation $I = \{p\}$), the update is equivalent to adding p to the speaker's commitment set. When there is more than one proposition in I (i.e., an information-seeking question), the speaker's commitment is just to the presupposition of the question (the disjunction of the alternatives that constitute the possible answers). Second, the denotation of the issue I is pushed on the stack. Usual stack operations are assumed; $\text{PUSH}(e, \tau)$ in (41b-ii) represents the new stack obtained by adding the issue e onto the stack τ .

3.3 Do not project: discourse conventions of *ba*

This section implements the proposal (repeated in 36) for *ba*-declaratives in a conversational scoreboard model.

(36) a. **Anti-Authority Condition:**

With a preajcent property P , *ba* presupposes that it is *not* common ground that the speaker is in a more epistemically privileged position concerning P .

b. **Weakened Addressee Compliance:**

Speaker of a *ba*-declarative with propositional content p does not assume that the Addressee will accept p as true in the future of the conversation.

As suggested before, the Anti-Authority Condition is implemented as a pragmatic presupposition triggered by *ba*, which restricts the input context in which a *ba*-declarative is felicitous. Following Zimmermann (2000) and Kaufmann (2012), I employ the concept of 'counting as an authority' in a context c , as defined in (37).

(37) x counts as an authority on a property P in context c iff

at all worlds w in the context set $cs_c = \bigcap cg$, it holds that

$$(\forall w' \in Bel_x(w))(\forall y) [w' \in P(y) \leftrightarrow w \in P(y)]. \quad (\text{adapted from Kaufmann 2012: 148})$$

(37) means that Speaker x has authority on a property P in c if, at every world w compatible with the context set, what she believes to be P (at w) is indeed P at w . $Bel_x(w)$ is a set of worlds compatible with what Speaker x believes at w . (37) is defined over properties, since the proposal will later be extended to capture interrogatives and imperatives. However, it can be straightforwardly simplified for cases where the semantic content of the preajcent is an issue (a set of propositions), as is needed to capture *ba*-declaratives.

- (38) x counts as an authority on an issue I in context c iff
 at all worlds w in the context set $cs_c = \bigcap CG$, it holds that
 $(\forall w' \in Bel_x(w))(\forall p \in I) [w' \in p \leftrightarrow w \in p]$.

Given (38), the Anti-Authority Condition is restated in (39).

(39) **Pragmatic presupposition for *ba* in declaratives**

With a prejacent denoting an issue I in context c , *ba* presupposes that it is not common ground that Speaker counts as an authority on I in c .

When to stop projecting. The special discourse effect conventionalized by *ba* is to stop projecting what the addressee should canonically respond. The question of when and how speakers project future discourse developments has been discussed by Rudin (2022). He proposes that the act of adding content to the Projected Set is subject to a set of pragmatic maxims, which, like the classical Gricean maxims, can be violated. Of particular interest to us here is the *Viability* maxim, introduced in (40).

(40) **VIABILITY (informal version):**

- a. Do not project a Common Ground if it is incompatible with some interlocutor's commitment;
- b. Do not project a Common Ground if you have reason to believe it is incompatible with some interlocutor's private beliefs. (Rudin 2022: 364)

According to (40), a projected common ground is viable only if it contradicts neither some interlocutor's commitments nor their private beliefs. Our proposal may therefore be restated as follows: *ba* is felicitous only if no resolution to the issue raised by the *ba*-utterance is viable.¹³ This generalization accounts for many contexts in which *ba* is felicitous, including disagreements and the interrogative cases to be discussed in §4. However, it is less clear how, in confirmation-seeking or addressee-uncertainty cases, a *ba*-utterance projects future common grounds that contradict the addressee's commitments or private beliefs. In fact, in confirmation-seeking cases, the speaker proposes a potential resolution precisely because she considers it to be viable. In such cases, the failure of a *ba*-utterance to project a compliant discourse move is due to the speaker's lack of epistemic privilege, rather than to any potential contradiction with the addressee. Therefore, rather than directly adopting the *Viability* maxim for *ba*, I propose a special updating convention for the Projected Set that remains neutral as to why the projection is blocked. A discourse move may fail to project its canonical discourse future not only because the projected future would contradict the addressee, but also because the speaker lacks the requisite authority to assume that the addressee will comply.

I propose that the weakening effect of *ba* should be characterized as a special discourse effect which modifies the basic *CD*ES of canonical assertions we introduced in (35). The intuition is that *ba* signals that the addressee accepting the move is not necessarily a future discourse projection. Therefore, the projected set (*PS*) stays the same as the input context after updating a *ba*-declarative. The discourse effects of *ba*-declaratives are summarized in (41).¹⁴

¹³I am indebted to Deniz Rudin (p.c.) for pointing it out to me.

¹⁴The *CD*ES associated with *ba*, particularly the update to the Projected Set, should be understood as a discourse consequence of *ba*'s Anti-Authority Condition rather than as a general operation on the Projected Set itself. In other words, the update rules proposed here are intended to illustrate how seemingly different pragmatic effects, such as weakening and strengthening, can be formalized in a unified way.

- (41) a. Special CDE of *ba*: the output projected set stays the same as the input.
 b. CDES of updating a *ba*-declarative *S* (content: *I*):
 $K_i + S = K_o$:
 i. $DC_{sp,o} = DC_{sp,i} \cup I$
 ii. $T_o = \text{PUSH}(I, T_i)$
 iii. $PS_o = PS_i$

Take (15) as an example, repeated in (42). Assume that before (42) is uttered, the input context has the following structure as in (43).

- (42) CONFIRMATION SEEKING (repeated from above)
 [The addressee is organizing a party. The speaker is wondering when exactly the party starts.]
 Jinwan paidui shi qidian **ba** .
 tonight party COP 7-o'clock BA
 ‘The party is at 7 o’clock tonight, right?’

- (43) K_i : Input context

$DC_{sp,i}$	TABLE	$DC_{a,i}$
$\{\dots\}$	$\langle \rangle$	$\{\dots\}$
COMMON GROUND CG_i		PROJECTED SET $PS_i = \{cg\}$

After (42) is uttered, both DC_{sp} and the Table are updated, while the projected set remains unchanged, as illustrated in (44).

- (44) K_o : Output context

$DC_{sp,o}$	TABLE	$DC_{a,o}$
$\{p, \dots\}$	$\langle \{p\} \rangle$	$\{\dots\}$
COMMON GROUND CG_o		PROJECTED SET $PS_o = PS_i = \{cg\}$

where $p = \text{The party is at 7 tonight.}$

To conclude, this section has formalized the Anti-Authority Condition and implemented the weakening of Addressee Compliance as a special discourse effect of *ba*, one that does not update the projected set with discourse future. In the next section, we consider how interrogatives complicate this account, discussing the puzzling ‘impatience’ effect when interacting with *ba*.

4 *ba* in interrogatives

This section aims to narrow down the contexts where *ba*-interrogatives are felicitous. *ba* is not felicitous for a question in truly out-of-the-blue scenarios. The *ba*-question in (45) is not felicitous, while the same *ba*-question becomes felicitous if the speaker A is impatient when she utters the question in (46).

- (45) [The teacher has just assigned a new set of problems and allows the students to work in pairs. A sees the first problem and asks her partner:]

A: Buhaoyisi, zhe ti zenme zuo (**#ba**)?
excuse-me this problem how do BA

A: 'Excuse me, (do you know) how to solve this problem?'

- (46) [A and B have been discussing how to solve a mathematical problem for a long time. B rejected every solution that A provided, A to B:]

A: Zhe ti zenme zuo **ba**?
this problem how do BA

'(Come on), how can this problem be solved?'

In (46), the scenario dubbed "impatience" by Ettinger and Malamud (2015), *ba*-questions typically carry a flavor of unfriendliness. After several unsuccessful attempts to resolve a math problem, the speaker becomes impatient and uses a *ba*-question to demand a solution. The contextual information that makes *ba* felicitous can be linguistic or extra-linguistic. For example, Speaker B in (47) is impatient because A goes back and forth on whether to dive. Although there is no immediate linguistic antecedent, a *ba*-interrogative can be used felicitously in (47):

- (47) [A sequence of people are waiting for diving. A stands in front of B, and A has been hesitating for a while.]

B: Ni tiao-bu-tiao **ba**
2SG jump-NEG-jump BA

'Are you going to jump or not?'

In a more explicit conversation like (48), the pre-jacent of the *ba*-interrogative appears in its preceding discourse. The effect that this *ba*-question brings is similar to the effect that *or-not* questions have in English: both types of questions 'force' the addressee to give an answer immediately. The question *will Zhangsan be fired or not* is also acceptable in (48).¹⁵

- (48) A: *Could you tell me if Zhangsan will be fired?*
Manager:...(irrelevant to the question)
A: *Could you answer the question directly, will Zhangsan be fired?*
Manager:...(still avoiding the question)

A: Zhangsan hui-bu-hui bei kaichu **ba**
Zhangsan will-NEG-will PASS fire BA

'Will Zhangsan be fired?'

Previous works have noticed the necessity of the speaker's impatience for *ba*-interrogatives to appear (Chao 1968; Han 1995; Ettinger and Malamud 2015). However, impatience alone is an insufficient criterion, as can be seen from the following modification of (46).

¹⁵The 'unfriendly' effect of *or-not* questions is also termed the *cornering* effect; for discussions particularly of the cornering effect, see Biezma (2009), Biezma and Rawlins (2012) and Biezma and Rawlins (2015).

- (49) [A is trying to solve a math problem of which B knows the solution.]
- i. A: Should I choose A?
 - ii. B: No.
 - iii. A: Hmm...Is B correct?
 - iv. B: No.
 - v. A: # C shi bu shi daan **ba**?
 C COP NEG is answer BA
 A: 'Is C the answer?'

(49) is a possible continuation of the conversation in (46). At the point where A is asking the question (49v), we can infer from the previous conversation that A has a reason for becoming impatient. But even though the impatience condition is satisfied, the *ba*-question (49v) is still infelicitous. The contrast between (46) and (49v) suggests that, rather than the speaker's impatience alone, *ba* is sensitive to the form and the content of the questions it attaches to. The infelicity of (49v) thus raises the question of what exactly the pragmatic conditions that license *ba*-interrogatives are, which we will further discuss in the rest of the section.

Another common position taken in the literature, also mentioned by Ettinger and Malamud (2015), is that a *ba*-interrogative can be regarded as a question embedded under an implicit imperative, as the *ba*-interrogative in (46) carries the meaning of '(tell me) how to solve the problem'. In this paper, I stand with Ettinger and Malamud (2015) in that this approach involves too many stipulations on the implicit structures, and there is no good explanation on why these implicit materials should be licensed here. Moreover, traditional speech act literature (e.g., Searle 1969) considers all kinds of interrogatives to be a subtype of directives, as does work on the Performative Hypothesis (e.g., Katz and Postal 1964; Ross 1970). However, none of these theories has predicted in any obvious way that all questions are 'impatient'.

Lastly, Han (1995) describes the meaning of *ba* in an interrogative sentence as 'I insist so', and suggests that *ba*-interrogatives are roughly equivalent to strong directives in the sense that *ba*-interrogatives indicate "the speaker's determination to get the hearer to perform a future action" (Han, 1995, 99). One argument Han gives is that for canonical questions without *ba*, the addressee has the freedom of giving all types of answers including 'I don't know'; whereas answers displaying the addressee's ignorance are not felicitous for *ba*-interrogatives. Han suggests that this is because a *ba*-interrogative is used as a directive to elicit a specific type of answer from the addressee. Although the observation Han (1995) describes about the felicitous type of answer is correct, I will show in the following sections that the constraints need to be made more precise.

4.1 (In-)felicitous uses of *ba*-interrogatives

Let us first consider the CAKE CASE in example (50). The scenario in (50) exemplifies a context where two kinds of *ba*-interrogatives are felicitous (50vi and 50vii). First, notice that in (50) the picky child B makes a public commitment at the beginning of the conversation uttering (50i); that is, there is some cake that B will eat. Building on (50i), the mother A then attempts to figure out the general question of *what cake B will eat* by asking two more specific questions (50ii) and (50iv). After receiving two negative answers from B, A loses her patience and uses a *ba*-question to reset the conversation: A can either ask the general question (i.e. the QUD) with *ba* explicitly (50vi), or a question challenging the

presupposition of the general question with *ba* as in (50vii). Note that the underlined parts in (50vi) and (50vii) signal that they are syntactically-marked interrogatives: *shenme* is a *wh*-word, meaning ‘what’; and *chi-bu-chi* is an A-not-A construction, which constitutes one type of polar question in Mandarin.

(50) THE CAKE CASE:

[A = Mother, B = picky child. The mother only prepares two kinds of cake for dinner: strawberry and chocolate.]

- i. B: *I want to eat cake.*
- ii. A: *Okay, will you eat strawberry cake?*
- iii. B: *No.*
- iv. A: *Will you eat chocolate cake?*
- v. B: *No.*
- vi. A: Ni chi shenme dangao **ba**?
2SG eat what cake BA
‘What cake will you eat (at all)?’
- vii. A: Ni chi-bu-chi dangao **ba**?
2SG eat-NEG-eat cake BA
‘Will you eat cake (after all)?’

Similar to the contrast between examples (46) and (49), the *ba*-question (51) cannot be felicitously used in the scenario in (50). That is, *ba* cannot attach to a question of the same type as the specific questions being asked in the previous conversation, such as (50ii) and (50iv).

(51) THE CAKE CASE (continued): A = Mother, B = Picky child

A”: #Ni chi-bu-chi nailao dangao **ba**?
2SG eat-NEG-eat cheese cake BA
‘Will you eat cheese cake?’

The question (52), which is the presupposition-challenging question without *ba* attached, is also infelicitous in the CAKE CASE. The reason of the infelicity of (52) is evident: with B’s explicitly stated commitment (50i), asking (52) ought simply to be redundant.

(52) THE CAKE CASE (continued): A = Mother, B = Picky child

A”: #Ni chi-bu-chi dangao?
2SG eat-NEG-eat cake
‘Will you eat cake?’

B’s commitment towards the presupposition of the question in the CAKE CASE is necessary for *ba*-questions to be felicitous. In a modified scenario in (53), which lacks the addressee B’s commitment, asking a *ba*-question to challenge the presupposition of the QUD becomes infelicitous since here ‘B eats cake’ is simply the speaker A’s implicit assumption.¹⁶

¹⁶Readers may notice that in the diving scenario (47), B’s utterance is at the beginning of the conversation and the addressee’s public commitment is absent; however, the *ba*-question is still felicitous. As I suggested above, the felicity conditions for *ba*-questions do not require explicit commitment; implicit commitments also count. In the diving scenario, A’s behavior (i.e. standing in the line of people waiting for diving) creates the implicit commitment that A intends to dive.

- (53) THE CAKE CASE (no commitment):
 [B is having dinner in A's house. A plans to serve some cake as dessert now.]
- i. A: *Will you eat strawberry cake?*
 - ii. B: *No.*
 - iii. A: *Will you eat chocolate cake?*
 - iv. B: *No.*
 - v. A: #Ni chi shenme dangao **ba**?
 2SG eat what cake BA
 'What cake will you eat (at all)?'
 - vi. A': #Ni chi-bu-chi dangao **ba**?
 2SG eat-NEG-eat cake BA
 'Will you eat cake (after all)?'

Before we make the conditions for *ba*-questions more precise in the next section, let us consider one more scenario to complete our outline of the felicitous contexts of *ba*-questions:

- (54) PREPARING FOR DINNER:
- i. B: *It's time to prepare for dinner.*
 - ii. A: *What do you want to eat?*
 - iii. B: *I don't eat meat, I don't eat vegetables and I don't eat fruit.*
 - iv. A: Ni xiang chi shenme **ba**?
 2SG want eat what BA
 'What do you want to eat?'
 - v. A': Ni chi-bu-chi wanfan **ba**?
 2SG eat-NEG-eat dinner BA
 'Are you going to eat dinner?'

In PREPARING FOR DINNER, all the contextual factors that *ba* is sensitive to are satisfied in the given scenario: the addressee B's commitment to the presupposition of the QUD (*What do you want to eat?*), but it does not seem that both interlocutors would reach a true answer at the point A utters the *ba*-question since B's utterance (54iii) nearly exhausts the space of possible answers. The exhaustion of nearly all the possible alternatives can also be found in the CAKE CASE: all kinds of cake the mother has on hand are strawberry and chocolate. This factor plays a crucial role in both scenarios: only when the speaker cannot think of any other possible alternatives can a *ba*-question be felicitously asked. The intuition behind this 'exhaustivity' condition is that asking a *ba*-question to challenge the validity of the QUD would be improper unless the previous discourse signals that the QUD is not answerable.

Moreover, a felicitous answer to the *ba*-question in PREPARING FOR DINNER must be a positive, 'true' answer. Any negative continuation along the lines of (54iii) is not acceptable in the scenario (54), as shown in (55).

- (55) PREPARING FOR DINNER (continued):
- a. B: *I don't eat meat, I don't eat vegetables and I don't eat fruit.*

- b. A: Ni xiang chi shenme **ba**?
 2SG want eat what BA
 ‘What do you want to eat?’
- c. B: #Wo ye bu chi mifan.
 1SG also NEG eat rice
 ‘I don’t eat rice either.’

The scenario PREPARING FOR DINNER again demonstrates that in addition to the ‘impatience’ condition, a more restrictive set of conditions for *ba*-questions is needed: a *ba*-interrogative is licensed only when the addressee commits to there being a solution to the issue, yet she does not steer the conversation towards a state where the issue is resolved. Moreover, there is a constraint on the host of *ba*: it must be the QUD. So, an account of *ba*-questions should also be able to explain *ba*’s sensitivity to the content of the question. In the next section I will show that the discourse leading to *ba*-questions requires a conversational crisis to exist in the context: the addressee has committed to the presupposition that the QUD is answerable, whereas the unsuccessful attempts at resolving the QUD in the previous conversation signal that the QUD might not be answerable. Rather than being the condition itself, the impatience is symptomatic of the actual underlying condition.

Interim summary. So far we have observed that (i) *ba* selects a particular type of questions, and (ii) a *ba*-marked question specifies two presuppositions for the input context. Preliminary generalizations are made in (56) and (57).

(56) **Descriptive generalizations on the content of *ba*-questions:**

Questions that *ba* can attach to:

- a. the current QUD;
- b. questions challenging the presupposition of the QUD.

(57) **Descriptive generalizations on discourse prior to *ba*-questions:**

A *ba*-interrogative can be used felicitously only if:

- a. the addressee has explicitly committed to the presupposition of the QUD;
- b. the previous conversation signals that the QUD has no true answer (or that the residual answer is true; Hamblin 1973).

5 *ba* strengthens Addressee Compliance in interrogatives

In this section will give an account of the behavior of *ba* in questions which contains two parts. The core proposal is that *ba* signals that one or more prior moves by the addressee have been non-compliant (e.g., avoiding a question or failing to address a question she should be able to), which results in a conversational crisis in the prior context. The strengthening effect of *ba* is a byproduct of the crisis. That is, *ba* is only felicitous in questions when the question itself seems unanswerable. First, I argue that *ba*-interrogatives traverse the discourse tree upward. This implies that the question to which *ba* attaches is either a discourse-old question that has already been placed on the Table, or a question that the speaker assumes the addressee has effectively already answered. Second, I propose two presuppositions on the input context of *ba*-interrogatives and show that the Anti-Authority Condition and special discourse effect of *ba* proposed for declaratives in §3 apply vacuously in interrogative contexts. §5.1 discusses how *ba*-interrogatives inform the hierarchical discourse structure.

§5.2 lays out the core proposal, the two presuppositions, and sample updates on the input contexts of *ba*-questions.

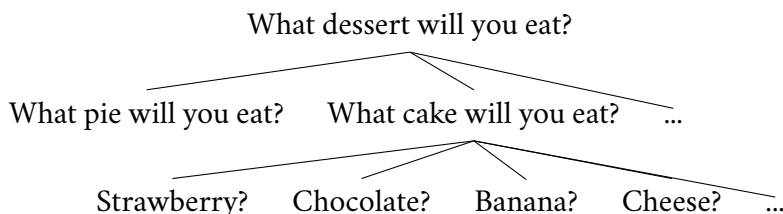
5.1 Mapping *ba*-interrogatives to discourse trees

As we have seen in §4, the type of questions *ba* may attach to is not random. It needs to be either a discourse-old question which interlocutors have been attempting to resolve, or a ‘vacuous’ question whose answer the speaker assumes is already in the common ground. In the former case, the question can be modeled as a QUD that both interlocutors have agreed to resolve and that has already been placed on the Table. The latter case is a parallel case to Stalnaker (1978)’s constraint against vacuous assertions: assertions cannot be made if they are already contextually entailed. In the interrogative context, this implies that a question cannot be asked if the true answer is entailed by the context. I suggest that attaching *ba* makes a vacuous questioning move felicitous because *ba* does not project any further updates to the common ground, thereby avoiding the potential updating conflict between the Projected Set and the common ground.

To model this intuition, I adopt the d-tree model (Roberts 1996; Büring 2003; Rojas-Esponda 2014) in which conversations are modeled as hierarchical structures of discourse moves. Each node in a d-tree represents a declarative or an interrogative sentence. These nodes can be understood either as questions set up for interlocutors to resolve or as answers to these questions.

I argue that *ba*-interrogatives traverse the discourse trees upward as follows. To resolve a complex question (the QUD), interlocutors in a conversation may proceed from the complex question to several subquestions, i.e., simpler questions providing complete or partial answers to the complex question. The QUD and its subquestions together form a discourse hierarchy, a d-tree, which contains a sequence of nodes of questions. (58) presents a possible d-tree of the conversation in THE CAKE CASE.

(58)



Take the CAKE CASE scenario in (50) as an example. The QUD of the conversation is *what cake will you eat*, which is brought up by B’s request (50i) *I want to eat cake*. By uttering (50i), the addressee makes sure that there should be at least some cake that she eats, which satisfies the commitment condition. Assuming this, the speaker uses several simpler subquestions (*Will you eat strawberry cake? Will you eat chocolate cake?*) as strategies to resolve the higher QUD. This is a natural flow within a d-tree, which goes from a higher node to lower nodes. However, as the conversation continues, the negative answers to the speaker’s subquestions collectively give the “residual” answer to the QUD (Hamblin 1973): there is in fact no true answer to it. In this situation, it is reasonable for the speaker to perform two kinds of moves using *ba*: either (i) the speaker assumes there should be no presupposition failure given the addressee’s commitment, and asks for the answer to the QUD by restating it; or (ii) the

speaker doubts whether the QUD is a valid conversational goal by challenging the presupposition of the QUD.

- (59) B: I want to eat cake. (QUD: **what cake will you eat?**)
 A: Okay, will you eat strawberry cake? (daughter question)
 B: No.
 A: Will you eat chocolate cake? (daughter question)
 B: No.
 A: What cake will you eat **ba**? (QUD)
 A': Will you eat cake **ba**? (Presupposition-challenging)

In sum, the standard traversal rule of a d-tree corresponds to the descending order of its nodes. Interlocutors may move from a node to its sister or daughter. I have argued that a *ba*-interrogative marks a move from a node to one of its predecessors. The content of a *ba*-question explicitly reflects either the higher-level question itself or the presupposition of its parent node.

5.2 Discourse dynamics of *ba* in interrogatives

This section formulates the presuppositions for *ba*-interrogatives. The presuppositions introduced below are not ad-hoc; they are specific realizations of the general requirement that a *ba*-utterance presupposes a potential or existing non-compliant move from the addressee. Previously, I have proposed the anti-Authority presupposition and the special discourse effect of *ba* in declaratives, repeated in (60).

- (60) a. **Anti-Authority Condition:** With a preadjacent highlighting property *P* in context *c*, *ba* presupposes that it is not common ground that Speaker counts as an authority on *P* in *c*.
 b. **Special CDE of *ba*:** the output projected set stays the same as the input.

(60a) is vacuously satisfied in an interrogative unless the question itself is a quiz question (or rhetorical in similar ways). (61) shows that *ba* is not felicitous in a quiz question.

- (61) Huida yixia wenti. #Zheli lambda shi shenme yisi **ba**?
 answer following question here lambda cop what meaning BA
 'Answer the following question. What does lambda mean here?'

(60b) explains the discourse effect of *ba*-interrogatives, just as it did for declaratives. When used to bring back a discourse-old QUD, the move made by a *ba*-interrogative is not to introduce a new issue onto the Table, but to restate an unresolved issue whose projected discourse future is already inconsistent. On the other hand, in a presupposition-challenging question, the answer is assumed to be already contextually entailed; therefore, no further updates to the common ground should be projected.

I assume that adding *ba* to an interrogative does not change the denotation of the question, (i.e., $\llbracket \phi ? \rrbracket = \llbracket \phi - ba ? \rrbracket$), but introduces presuppositions on the input contexts. In (62) *cg* represents a future common ground in the ps, and Q^+ represents the QUD (or the issue).

- (62) Formally, *ba* adds the following two presuppositions:
 a. $\lambda w. \exists p \in Q^+[p(w)] \in \text{DC}_{a,i}$

$$b. \text{ ps}_i \cup Q^+ = \emptyset \text{ (i.e. for all } cg \in \text{ps}_i, \bigcap (cg \cup \{Q^+\}) = \emptyset)$$

When (62a) and (62b) are satisfied, $\phi - ba?$ is felicitous only if $\llbracket \phi \rrbracket = Q^+$, or $\llbracket \phi \rrbracket = \{Q^+, \bigcup Q^+\}$

(62a) accounts for our first generalization (56). The formula in (62a) tells us that the proposition of there being a true answer to the Q^+ is in the addressee's input commitment set. (62b) sets a condition on the input ps such that updating the ps with the QUD Q^+ returns an empty set. In other words, all the future common grounds in the input ps for a ba -question should be inconsistent. It amounts to saying that the issue on the Table is not resolvable. This accounts for our second observation: before uttering ba -interrogatives the previous discourse entails that its QUD (Q^+) is not answerable.

Let us see how (62) implements our informal generalizations of the meaning contributions made by ba . Take conversation (50) again as an example. Before anything is uttered, we assume that the initial context state K_1 has the following structure:

(63) K_1 : Initial context state

DC_{sp}	TABLE	DC_a
$\{\dots\}$	$\langle \rangle$	$\{\dots\}$
COMMON GROUND cg		PROJECTED SET $\text{ps} = \{cg\}$

When (50i) *I want to eat cake* is uttered, the addressee is committed to there being a true answer to the QUD *what cake will you eat*. In other words, the addressee has committed herself to the proposition in the form of $p_1 \vee p_2$, assuming that only strawberry cake (p_1) and chocolate cake (p_2) count as possible answers to the QUD. When the speaker says *okay* in (50ii) and starts looking for an answer to the QUD, we assume that at this point both interlocutors agree that they are going to resolve this issue, and thus the QUD *what cake will you eat* is introduced and put onto the Table.

(64) K_2 : The context state after *I want to eat cake*.

DC_{sp}	TABLE	DC_a
$\{\dots\}$	$\langle \{p_1, p_2\}, \langle \rangle \rangle$	$\{p_1 \vee p_2, \dots\}$
$cg = \{p_1 \vee p_2, \dots\}$		$\text{ps} = \{cg \cup \{p_1\}, cg \cup \{p_2\}\}$

where $p_1 = a$ will eat strawberry cake, $p_2 = a$ will eat chocolate cake.

Next, the speaker asks a subquestion of the QUD, *will you eat strawberry cake*. The addressee, by answering with *no*, has committed herself to the proposition $\neg p_1$. Hence the DC_a , the cg , and the ps are all updated with the proposition $\neg p_1$. We see that one of the future cgs become inconsistent after updating with $\neg p_1$, and is thus discarded. The addressee's answer completely resolves the subquestion, and thus the subquestion is popped off the stack. (65) shows the context state after updating with the addressee's answer *no* in (50iii).

(65) K_3 : The context state after -*Will you eat strawberry cake?* -*No*.

DC_{sp}	TABLE	DC_a
$\{\dots\}$	$\langle\{p_1, p_2\}, \langle\rangle\rangle$	$\{p_1 \vee p_2, \neg p_1, \dots\}$
$cg = \{p_1 \vee p_2, \neg p_1, \dots\}$	$ps = \{cg \cup \{p_1\} \cup \{\neg p_1\}, cg \cup \{p_2\} \cup \{\neg p_1\}\}$	

$p_1 = a$ will eat strawberry cake

The update with the second subquestion *will you eat chocolate cake* and its response *no* repeats the process in (65) with p_2 . After updating with the addressee's commitment $\neg p_2$, the remaining future cg in the ps is inconsistent and is thus eliminated, the ps becomes empty, which satisfies the presupposition (62b), the input $ps \sqcup Q^+ = \emptyset$. The ps turns out to be empty, but the QUD is still on top of the Table, which gives us a conversational crisis.

(66) K_4 : The context state after -*Will you eat chocolate cake?* -*No*.

DC_{sp}	TABLE	DC_a
$\{\dots\}$	$\langle\{p_1, p_2\}, \langle\rangle\rangle$	$\{p_1 \vee p_2, \neg p_1, \neg p_2, \dots\}$
$cg = \{p_1 \vee p_2, \neg p_1, \neg p_2, \dots\}$	$ps = \{cg \cup \{p_2\} \cup \{\neg p_1\} \cup \{\neg p_2\}\}$	

$p_2 = a$ will eat chocolate cake.

Notice also that the addressee's commitment of there being a true answer to the QUD is in the DC_a , which satisfies the presuppositions (62a). When the input presuppositions are all satisfied, both *ba*-questions *what cake will you eat ba* (QUD restated) and *will you eat cake ba* (presupposition of the QUD) are possible moves.

6 Implications on the imperative uses of *ba*

So far, the theory of *ba* put forward in this paper holds that its opposite discourse effects in declaratives and interrogatives are derived by constraining the projected set. Following from this, it would be natural to ask what it means for other clause types such as an imperative to *project* its future discourse development. Farkas and Bruce (2010) do not discuss how imperatives project future discourse developments within the Table model themselves. Nevertheless, the issue of what discourse future imperatives project has received considerable attention in the literature (see, e.g., Portner 2004; Lauer 2013; Rudin 2018; Starr 2018; Scheffler and Malamud 2023; Penka and Romero 2025). Empirically, extending the current proposal of *ba* to imperatives is a question that cannot be overlooked, since *ba* has long been described as an imperative marker in the descriptive literature (e.g., Chao 1968; Zhu 1982; Po-Ching and Rimmington 2015). This section first discusses the implications conveyed by imperatives and the use of *ba* in imperatives. I then demonstrate that the current theory of *ba* naturally extends to imperatives and correctly predicts the distribution of *ba* in imperatives.

6.1 Imperatives and *ba*

Imperatives have long been a challenge for standard formal semantic frameworks because they exhibit a wide range of uses, a phenomenon known as the *problem of functional heterogeneity* (Schmerling 1982). Finding a unifying property that characterizes all imperative uses has proven difficult. However, despite their wide usage, perhaps the central intuition underlying the most stereotypical uses of imperatives, such as commands and orders (e.g., *Leave!*), is that they are used to get the addressee to perform an action. The implication that the addressee brings about the indicated action is termed *addressee inducement* by Condoravdi and Lauer (2012), a notion that will be useful later in characterizing *ba*-imperatives.¹⁷ In cases where there is no addressee, or where the realization of the imperative is beyond the addressee's control (e.g., the wish uses such as *Get well soon!*), the imperative still conveys that the speaker wants the imperative to be realized, even though neither interlocutor is in a position to actualize it. The overall effects of stereotypical imperatives such as commands or orders are summarized in (67).

- (67) *Leave!*
 (I) expresses: The addressee leaves; (content)
 (II) conveys that the speaker wants the addressee to leave; (speaker desire)
 (III) acts as an inducement for the addressee to leave. (addressee inducement)
- (Condoravdi and Lauer 2012: 38)

Depending on whether speaker desire and addressee inducement are present, I classify imperative uses into four groups, following Condoravdi and Lauer (2012) and Kaufmann (2012): directives, wish-type uses, permissions and invitations, and advice, as illustrated in (68). The discussion of the imperative uses of *ba* below follows this classification.¹⁸

- (68) a. **Group I:** directives (*Leave!*) speaker desire: ✓, addressee inducement ✓
 b. **Group II:** wish-type uses (*Get well soon!*) speaker desire: ✓, addressee inducement: ✗

¹⁷As it is beyond the scope of this paper to do justice to the extensive literature on imperatives, I follow the descriptive generalization of the implications of imperatives proposed by Condoravdi and Lauer (2012) and Lauer (2013) as a working approximation. Various accounts have been proposed to capture the distributional and functional profile of imperative clauses, summarized into roughly two streams:

1. Modality as part of the conventional meaning (Aloni 2007; Kaufmann 2012, 2019; Condoravdi and Lauer 2012 a.o.): this line of research treats imperatives semantically on a par with modalized declaratives, with the assumption that there is a propositional operator in the logical form.
2. Non-representational accounts (Portner 2004; Von Stechow 2017; Starr 2020 a.o.): on this view, the discourse function of imperatives is not to represent the world; the directive meaning is achieved by updating distinct discourse components (e.g., *To-Do list* for Portner, *Preferences* for Starr).

¹⁸Condoravdi and Lauer (2012)'s original classification treats *advice* with a disinterested speaker as a separate group, termed *disinterested advice*. Disinterested advice is special for their purposes because, unlike other types of advice, it appears to convey neither *speaker desire* nor *addressee inducement*. In giving disinterested advice such as *Take the red line!*, the speaker does not actively want the realization of the content since she is not interested in the addressee's future behavior. But she cannot stand against the realization of the content either. By contrast, advice given by a doctor to a patient (*Take these pills three times a day!*), for example, conveys both implications. For the purposes of the present paper, I follow Kaufmann (2012) in grouping the two types together. I argue that the felicity of *ba* in advice uses depends simply on whether the speaker is in an (epistemically) authoritative position.

- c. **Group III:** permissions and invitations (*Okay, go!*)
speaker desire: **X**, addressee inducement: **X**
- d. **Group IV:** advice (*Take these pills three times a day!*)
speaker desire: **?**, addressee inducement: **?**

Not all of the imperative uses in (68) can be realized with *ba*. Similar to its use in declaratives, *ba* has been claimed to weaken the force of an imperative. In the descriptive literature, Chao (1968) characterizes *ba* as an “advisative particle”. Similarly, Po-Ching and Rimmington (2015) notes that *ba* is “...commonly used to make suggestions” (Po-Ching and Rimmington 2015: 409). Recently, the effect of *ba* in imperatives has been characterized as “softening the speaker’s tone of voice since a bare imperative gives a command that sounds harsher than a negotiation” (Deng 2015: 37-38; see also Ettinger and Malamud 2015). An example of *ba* turning a command into a suggestion is illustrated in (69).

- (69) a. Hui jia!
 return home
 ‘Go home!’
 b. Hui jia **ba**.
 return home BA
 ‘How about going home?’/ ‘Let’s go home.’ (Deng 2015: 38)

When the context clearly specifies that the imperative is intended as a command with a forceful tone, *ba* is infelicitous, as shown in (70).

- (70) [The speaker is angry with her child for staying outside too long.]
 Xianzai like hui jia {Ø/ #ba}!
 Now immediately return home BA
 ‘Go home now!’

Examples (69) and (70) illustrate the basic restrictions that *ba* imposes on directive uses of imperatives. The next subsection examines the restrictions on *ba* in the remaining groups of imperative uses and shows that the proposed theory of *ba* correctly predicts its distribution in imperatives.

6.2 Imperatives without obligations

We have proposed that *ba* signals potential or existing non-compliant discourse moves from the addressee. The particular addressee move that counts as non-compliant depends on the speech act to which it responds. At the level of discourse dynamics, I have given these cases a unified treatment: *ba* does not project (compliant) future discourse updates. Given this proposal, what predictions does the current theory of *ba* make for imperative uses? The first issue to tackle is what Addressee Compliance amounts to in response to a stereotypical imperative, namely a directive. As discussed in §6.1, the intuition is captured by the notion of *addressee inducement*: in response to a directive, compliant discourse moves involve the addressee carrying out the indicated action. Such a discourse move may be realized silently without any linguistic cues, or overtly through a response such as *Will do*. A working definition of Addressee Compliance for canonical directives is introduced in (71).

(71) **Addressee compliance for canonical directives:**

Speaker presents herself as assuming that Addressee will carry out the indicated action in the future of the conversation.

(71) gives rise to one straightforward prediction and one complication. First, imperative uses that lack *addressee inducement* should not project such a compliant discourse move. As we will see below, this prediction is borne out by the distribution of *ba*. Second, the presence of *addressee inducement* does not by itself guarantee the projection of a compliant discourse future. For instance, a plea (a subtype of directives) such as *Please, lend me the money!* is intended to induce the addressee to perform the requested action. Nevertheless, it would be too strong for the speaker to project that the addressee will in fact lend her the money, since the speaker lacks the authority to obligate the addressee to comply.¹⁹ This suggests the connection between speaker authority and the projected discourse future also holds in imperatives. A compliant move by the addressee is projected only when the speaker has authority over the addressee such that the addressee is obligated to actualize the imperative. In other words, only the canonical subset of directives (commands or orders) that create an obligation project a discourse future in which the addressee performs the indicated action. We will see in the next section that *ba* is generally infelicitous in directive uses, but acceptable in certain less authoritative directives.

6.2.1 Predictions

Speaker authority typically gives rise to addressee obligation. But the Anti-Authority Condition imposed by *ba* confines its use to contexts in which the imperative does not create such an obligation. That is, the discourse effect of *ba* in imperatives is to block the projection of a future discourse in which the addressee fulfills the imperative. The most immediate prediction is that *ba* should be felicitous in Groups II and III, where *addressee inducement* is not present at all. This is borne out as shown in (72).²⁰

(72) a. **Group II:** wish-type uses

i. Kuaidian hao-qilai { $\emptyset$ /**ba**}!

quickly good-up BA

‘Get well soon!’

(well-wish: ✓)

ii. Qu si { $\emptyset$ /**ba**}!

go die BA

‘Go die!’

(curse: ✓)

iii. Xia-dian yu { $\emptyset$ /**ba**}!

rain-a.bit rain BA

‘(Please,) rain a bit!’

(addressee-less wish: ✓)

b. **Group III:** permissions and invitations

¹⁹In the discussion of imperatives, I broaden the concept of authority from epistemic authority to include other forms of authority, such as social authority. How inferences about the speaker’s social background should be incorporated into a semantic or pragmatic theory is beyond the scope of this paper; see, for example, Beltrama and Papafragou (2023) for related discussion.

²⁰Absent-wish uses with individual-level predicates, such as *Be tall!*, are generally impossible in Mandarin. Similar meaning can instead be conveyed with stage-level predicates, as in *Lai yi-ge gao-de ba!* ‘Please, let a tall person come!’

- i. Hao, qu wan { $\emptyset$ / **ba**}!
okay go play BA
‘Okay, go play!’ (permission/concession: ✓)
- ii. He-dian shui { $\emptyset$ / **ba**}!
drink-a.bit water BA
‘Drink some water(, if you like)’ (offer: ✓)
- iii. Wanshang lai chifan { $\emptyset$ / **ba**}!
evening come eat BA
‘Come and have dinner tonight!’ (invitation: ✓)

A commonality shared by Group II and Group III uses is that these imperatives are less authoritative. In wish uses, neither the speaker nor the addressee can help realizing the content. In permissions or invitations, whether the imperative is fulfilled depends entirely on the addressee’s desires. Although the speaker permits or welcomes the realization of the imperative, the addressee is under no obligation to carry it out.²¹

In Group I directive uses, *ba* is infelicitous with commands/orders and warnings, and acceptable with requests and pleas where the speaker lacks the requisite authority, as in (73).²² This variation across contexts is expected, as it should be if *ba*’s felicity is about speaker authority.

(73) **Group I: directives**

- a. [A firefighter directing people to leave a burning building]
Kuai likai zher { $\emptyset$ / #**ba**}!
quickly leave here BA
‘Leave here now!’ (command/order: ✗)
- b. Bie dong { $\emptyset$ / #**ba**}!
PROH move BA
‘Don’t move!’ (warning: ✗)
- c. Na-dian yan lai { $\emptyset$ / **ba**}!
hand-a.bit salt come BA
‘Hand me some salt(, please!)’ (request: ✓)
- d. Jie wo dian qian { $\emptyset$ / **ba**}!
lend 1sg a.bit money BA
‘Lend me some money(, please!)’ (plea: ✓)

Lastly, Group IV advice uses require separate treatment because utterances used as advice may be ambiguous between an assertive speech act that simply presents what the best solution is and a directive speech act that induces the addressee to perform the action (cf. Searle and Vanderveken 1985). This issue is particularly challenging in Mandarin because imperatives are generally not morphosyntactically distinct from declaratives. The same utterance in Mandarin can be used as a directive to get

²¹In permission uses, there is always a pre-existing proposal initiated by the addressee. The permission functions as an acceptance of that proposal, rather than projecting any further discourse development.

²²See Condoravdi and Lauer (2010) for a related discussion on the speaker authority distinction between commands/warnings and requests/pleas.

the addressee to do something, or a fragment answer providing the optimal solution. This ambiguous interpretations merit closer investigation.

Advice uses are generally incompatible with *ba*. Giving advice places the speaker in an epistemically privileged position, or at least requires the speaker to act as if they occupy one. In directive advice uses, such as a doctor directing a patient to take medication, *ba* is infelicitous, as shown in (74).

- (74) Zhe-ge yao yi-tian chi san-ci { $\emptyset$ / #**ba**}.
 this-CL medicine one-day eat three-time BA
 ‘Take this medicine three times a day.’ (directive-type advice: ✗)

In the doctor-patient relationship, the doctor (the speaker) occupies both a socially and epistemically authoritative position. In any unmarked context, the patient has no obvious reason to doubt or suspect that the doctor is being untruthful. Since the doctor’s directive imposes an obligation on the patient to comply, *ba* is predicted to be infelicitous.

In a less straightforward case, such as (75), where speaker A is asking for directions, *ba* is infelicitous in the advice speaker B offers.

- (75) [Strangers in the streets.]
 A: Buhaoyisi, qing wen kafeiting zenme qu?
 excuse.me please ask cafe how go
 ‘Excuse me, How do I get to a cafe?’
 B: Qianmian zuo zhuan { $\emptyset$ / #**ba**}.
 ahead left turn BA
 ‘Turn left ahead.’

The context in (75) presupposes that speaker B is in a more epistemically authoritative position than A, or at least that she presents herself as occupying one when giving directions. In this context, there is also no reason to doubt that speaker A will follow speaker B’s advice, thus projecting a future discourse move in which the advice is carried out.

ba can be felicitous when B is uncertain whether her answer is the true answer to A’s question, as shown in (76). In this case, B’s utterance is assertive rather than directive: B uses *ba* to offer the best answer she can, rather than to direct A’s actions.

- (76) A: Qing wen kafeiting zenme qu?
 please ask cafe how go
 ‘How do I get to a cafe?’
 B: Qianmian zuo zhuan **ba**... wo ye bu zhidao.
 ahead left turn BA 1SG also NEG.IMP know
 ‘(Maybe) turn left ahead...I don’t know either.’

The distinction between assertive and directive force can be diagnosed using the *That's (not) true!* test: unlike assertions, imperatives reject responses judging the truth values of the utterances, as illustrated in (77). The *ba*-utterance in (76) can be affirmed or denied by *That's (not) true!*, as in (78).²³

(77) A: Close the window!

B: # That's true!/ # That's not true!

(78) A: kafeiting zenme qu?

cafe how go

'How do I get to a cafe?'

B: Qianmian zuo zhuan **ba**...

ahead left turn BA

'(Maybe) turn left ahead...'

A: O, ni shuo-de dui!/ Bu dui, qianmian zuo zhuan shi jianshenfang.

oh 2SG say-REL right NEG.IMP right ahead left turn COP gym

'Oh, you are right!/No that's not right, (if you) turn left ahead, it's a gym.'

To conclude this section, we have seen that *ba* exhibits a weakening effect in imperative uses that is identical to its effect in declaratives. The weakening effect in imperatives is reflected in the fact that *ba* is felicitous only in imperative uses that do not impose an obligation on the addressee to comply, including wish-type uses, permissions, requests, and pleas. Stereotypical imperative uses, such as directives and directive-type advice, are incompatible with *ba*. This distribution of *ba* follows directly from the present proposal, according to which *ba* is anti-Authority and stops projecting a compliant discourse future of the utterance.

Our discussion has not addressed the question of how to formalize a compliant projected discourse move of a directive within a discourse model, although the extensive formal literature on imperatives offers several promising approaches. For instance, Portner (2007) introduces a dedicated discourse component, the *To-Do List*, to represent the discourse effects of imperatives. Under this approach, the projected compliant discourse move of a directive can be modeled as adding the imperative content to the addressee's To-Do List for her to carry it out. Rudin (2018) extends the Table model to imperatives by introducing a set of teleological discourse components parallel to their doxastic counterparts, including a teleological projected set. The discourse effects of a directive can thus be updated in the teleological system. More recently, Scheffler and Malamud (2023) argue that the issue placed on the Table by an imperative is *will p*, which also projects *will p* as a potential discourse update. Following this approach, our analysis of *ba* can be naturally extended by treating it as blocking the projection of *will p*. A detailed comparison of these approaches, as well as a formal analysis of imperatives within the Table model, is left for future research.

²³Directive advice uses such as (74) and (75) generally pass the positive version of the *That's (not) true!* test (e.g., *That's true!* or *You're right!*), but reject the negative version (e.g., *That's not true!* or *You're wrong!*). This asymmetry is discussed by Kaufmann (2012) particularly for advice uses (Kaufmann 2012: 166).

7 Conclusion

This paper has shown, through a study of the Mandarin discourse particle *ba*, how constraining the projected course of discourse can have the opposing effects of either weakening or strengthening the Addressee Compliance assumption, depending on the clause type in which it is used. In the proposal developed here, *ba* signals potential or existing non-compliant discourse moves from the addressee. This intuition is captured by proposing that the discourse effect of *ba* is not to project updates to future common grounds. In declaratives, not projecting the addressee’s acceptance amounts to assuming the possibility of a non-compliant move from the addressee, which generates a weakening effect. In interrogatives, not projecting any possible answers to the question amounts to assuming a conversational crisis: the question itself seems unanswerable. Restating a seemingly unanswerable question with *ba* strengthens its interrogative force. The proposal is implemented in a Farkas and Bruce (2010)-style Table model, and has been shown to successfully capture the weakening effect of *ba* in imperative uses. I conclude that *ba* presupposes an epistemic asymmetry between the speaker and addressee in the input context, and contributes a special discourse effect that keeps the output projected set the same as the input.

The current proposal also contributes to the discussion of formal models of discourse dynamics. I showed that the projected component in the Table model is indeed necessary for modeling non-canonical speech acts. Various dimensions of meaning are also needed to model the uses of *ba*: the notion of QUD and the Lewisian conversational scoreboard. If discourse particles are to mark special conversational moves, studying discourse particles helps us explore in-depth general principles of pragmatics, and sheds light on modeling the dynamics of conversation in a more fine-grained way. Explaining the pragmatic effects of *ba*, polarized as they are across different clause types, underscores the need for discourse models that predict the outlook of a conversation and characterize conventional meaning of discourse future.

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